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import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

X = np.array([
    [2015, 50000, 1.2],
    [2016, 45000, 1.5],
    [2017, 40000, 1.5],
    [2018, 35000, 1.6],
    [2019, 30000, 1.5],
    [2020, 25000, 1.2],
    [2021, 20000, 1.5],
    [2022, 15000, 1.6],
    [2023, 10000, 1.5],
    [2024, 5000, 1.2],
    [2014, 60000, 1.4],
    [2017, 42000, 1.3],
    [2019, 28000, 1.5],
    [2021, 18000, 1.6],
    [2023, 8000, 1.2]
])

y = np.array([
    450000, 500000, 550000, 650000, 700000,
    800000, 900000, 1000000, 1100000, 1200000,
    400000, 530000, 720000, 920000, 1150000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("CAR PRICE PREDICTION")
print("-----")

print("\nActual Prices:")
print(y_test)

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print("\nPredicted Prices:")
print(y_pred)

print("\nMAE:", mean_absolute_error(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

new_car = np.array([[2022, 12000, 1.5]])

prediction = model.predict(new_car)

print("\nPredicted Price for New Car:")
print(round(prediction[0], 2))
```

... CAR PRICE PREDICTION

Actual Prices:

[1200000 530000 450000]

Predicted Prices:

[1206989.92636889 605404.53904152 414114.11578912]

MAE: 39426.78320709864

MSE: 2340833421.434512

R2 Score: 0.9793009424318033

Predicted Price for New Car:

974785.27